

```
In [4]: import pandas as pd
data = pd.read_csv("D:\\Data Engineering\\Python\\car_data.csv")
```

```
In [ ]: #Different ways of Filtering
```

```
In [ ]: #Method 1 - Petrol cars with all columns    normal method
```

```
In [5]: data.head()
```

```
Out[5]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Tran
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	

```
In [6]: data[data['Fuel_Type'] == "Petrol"].head(5)
```

```
Out[6]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Tran
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	
6	ciaz	2015	6.75	8.12	18796	Petrol	Dealer	
10	alto 800	2017	2.85	3.60	2135	Petrol	Dealer	

```
In [ ]: #Method 1.1 - Petrol cars with selected columns    normal method
```

```
In [7]: data[data['Fuel_Type'] == "Petrol"][["Car_Name", "Year", "Kms_Driven"]].head
```

```
Out[7]:
```

	Car_Name	Year	Kms_Driven
0	ritz	2014	27000
2	ciaz	2017	6900
3	wagon r	2011	5200
6	ciaz	2015	18796
10	alto 800	2017	2135

```
In [ ]: #Method 2 - Petrol cars with all columns      query method
```

```
In [8]: data.query('Fuel_Type == "Petrol"]').head()
```

```
Out[8]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Tra
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	
6	ciaz	2015	6.75	8.12	18796	Petrol	Dealer	
10	alto 800	2017	2.85	3.60	2135	Petrol	Dealer	



```
In [ ]: #Method 2.1 - Petrol cars with selected columns      query method
```

```
In [9]: data.query('Year == 2015')[["Car_Name", "Selling_Price", "Kms_Driven"]].head()
```

```
Out[9]:
```

	Car_Name	Selling_Price	Kms_Driven
6	ciaz	6.75	18796
7	s cross	6.50	33429
9	ciaz	7.45	42367
11	ciaz	6.85	51000
12	ciaz	7.50	15000

```
In [ ]: #Method 3 - Petrol cars with all columns      loc method (loc and iloc)
```

```
In [11]: data.loc[data["Fuel_Type"] == "Petrol"].head()
```

```
Out[11]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Tra
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	
6	ciaz	2015	6.75	8.12	18796	Petrol	Dealer	
10	alto 800	2017	2.85	3.60	2135	Petrol	Dealer	



```
In [ ]: #Method 3.1 - Petrol cars with selected columns      loc method (loc and i
```

```
In [12]: data.loc[data["Year"] == 2015][["Car_Name", "Selling_Price", "Kms_Driven", "Fuel_Type"]]
```

```
Out[12]:
```

	Car_Name	Selling_Price	Kms_Driven	Fuel_Type
6	ciaz	6.75	18796	Petrol
7	s cross	6.50	33429	Diesel
9	ciaz	7.45	42367	Diesel
11	ciaz	6.85	51000	Diesel
12	ciaz	7.50	15000	Petrol

```
In [ ]: #Filtering with multiple field with all columns
```

```
In [15]: data.loc[(data['Fuel_Type'] == "Petrol") & (data["Transmission"] == 'Autom')]
```

```
Out[15]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission
12	ciaz	2015	7.50	9.94	15000	Petrol	Dealer	Manual
22	sx4	2011	4.40	8.01	50000	Petrol	Dealer	Manual
40	baleno	2016	5.85	7.87	24524	Petrol	Dealer	Manual
45	ciaz	2014	7.50	12.04	15000	Petrol	Dealer	Manual
49	ciaz	2017	7.75	9.29	37000	Petrol	Dealer	Manual

```
In [ ]: #Filtering with multiple field with selected columns loc method
```

```
In [16]: data.loc[(data['Year'] == 2016) & (data['Fuel_Type'] == "Petrol") & (data['Seller_Type'] == 'Dealer')]
```

```
Out[16]:
```

	Car_Name	Selling_Price
40	baleno	5.85
165	Activa 3g	0.45
177	Honda Activa 125	0.35
275	city	10.90
285	jazz	7.40

```
In [ ]: #Filtering with multiple field with selected columns nomral method
```

```
In [17]: data[(data["Year"] == 2017) & (data["Seller_Type"] == 'Dealer')][["Car_Nam
```

```
Out[17]:
```

	Car_Name	Selling_Price	Present_Price
2	ciaz	7.25	9.85
10	alto 800	2.85	3.60
21	ignis	4.90	5.71
49	ciaz	7.75	9.29
52	innova	18.00	19.77

```
In [ ]: #Filtering with multiple field with selected columns query method
```

```
In [41]: data.query('Fuel_Type == "Petrol" and Transmission == "Automatic")["Car_
```

```
Out[41]:
```

	Car_Name	Selling_Price	Year	Present_Price
12	ciaz	7.50	2015	9.94
22	sx4	4.40	2011	8.01
40	baleno	5.85	2016	7.87
45	ciaz	7.50	2014	12.04
49	ciaz	7.75	2017	9.29

```
In [43]: data.query('Fuel_Type == "Petrol" and Transmission == "Automatic")["Car_
```

```
Out[43]:
```

	Car_Name	Fuel_Type	Transmission
12	ciaz	Petrol	Automatic
22	sx4	Petrol	Automatic
40	baleno	Petrol	Automatic
45	ciaz	Petrol	Automatic
49	ciaz	Petrol	Automatic

```
In [ ]: #Sorting the values
```

```
In [47]: data.sort_values("Selling_Price",ascending = False).head()
```

Out[47]:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Tra
86	land cruiser	2010	35.0	92.60	78000	Diesel	Dealer	
64	fortuner	2017	33.0	36.23	6000	Diesel	Dealer	
63	fortuner	2015	23.5	35.96	47000	Diesel	Dealer	
82	innova	2017	23.0	25.39	15000	Diesel	Dealer	
51	fortuner	2015	23.0	30.61	40000	Diesel	Dealer	

```
In [48]: data.sort_values("Kms_Driven",ascending = False).head()
```

Out[48]:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Ti
196	Activa 3g	2008	0.17	0.52	500000	Petrol	Individual	
179	Honda Karizma	2010	0.31	1.05	213000	Petrol	Individual	
84	innova	2005	3.49	13.46	197176	Diesel	Dealer	
85	camry	2006	2.50	23.73	142000	Petrol	Individual	
77	corolla	2004	1.50	12.35	135154	Petrol	Dealer	

```
In [ ]: #Question to be answered: (Selling price : asc, Kms_Driven : desc)
```

```
In [57]: data.sort_values(["Selling_Price","Kms_Driven"],ascending = [True,False]).
```

Out[57]:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Ti
200	Bajaj Pulsar 150	2006	0.10	0.75	92233	Petrol	Individual	
199	Honda CB Shine	2007	0.12	0.58	53000	Petrol	Individual	
198	Bajaj Discover 125	2011	0.15	0.57	35000	Petrol	Individual	
197	Honda CB twister	2010	0.16	0.51	33000	Petrol	Individual	
196	Activa 3g	2008	0.17	0.52	500000	Petrol	Individual	
195	Bajaj ct 100	2015	0.18	0.32	35000	Petrol	Individual	
190	Bajaj Pulsar 150	2008	0.20	0.75	60000	Petrol	Individual	

```
In [61]: data[data['Year'] == 2015]["Car_Name"].count() #Counting rows
```

Out[61]: 61

```
In [62]: data[data['Year'] == 2017]["Car_Name"].count()
```

Out[62]: 35

```
In [74]: data.groupby('Year')[["Car_Name"]].count().sort_values('Car_Name',ascending=True)
```

Out[74]:

Car_Name	
Year	
2015	61
2016	50
2014	38
2017	35
2013	33
2012	23
2011	19
2010	15
2008	7
2009	6
2005	4
2006	4
2003	2
2007	2
2004	1
2018	1

```
In [75]: data.head(1)
```

Out[75]:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transaction_Type
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Warranty



```
In [81]: data.groupby(["Year", "Fuel_Type", "Transmission"])[["Car_Name"]].count() #C
```

Out[81]:

Car_Name			
Year	Fuel_Type	Transmission	
2003	Petrol	Manual	2
2004	Petrol	Automatic	1
2005	Diesel	Manual	1
	Petrol	Manual	3
		Automatic	1
2006	Petrol	Manual	3
		Automatic	2
2007	Petrol	Manual	2
2008		Automatic	3
	Petrol	Manual	4
		Automatic	1
2009	Petrol	Manual	5
		Automatic	1
	Diesel	Manual	4
2010		Automatic	2
	Petrol	Manual	9
	CNG	Manual	1
2011	Diesel	Manual	1
		Automatic	1
	Petrol	Manual	16
2012		Automatic	2
	Diesel	Manual	5
	Petrol	Manual	16
2013		Automatic	1
	Diesel	Manual	4
		Automatic	2
2014	Petrol	Manual	26
		Automatic	2
	Diesel	Manual	12
2015		Automatic	2
	Petrol	Manual	22
	CNG	Manual	1
2016		Automatic	3
	Diesel	Manual	12
		Automatic	5
2017	Petrol	Manual	40
		Automatic	1
	Diesel	Manual	7
2018		Automatic	5
	Petrol	Manual	37

Car_Name			
Year	Fuel_Type	Transmission	
2017	Diesel	Automatic	3
		Manual	1
	Petrol	Automatic	5
		Manual	26
2018	Diesel	Manual	1

```
In [82]: import pandas as pd
df = pd.read_excel("D:\\Data Engineering\\Python\\Sample - Superstore.xlsx")
```

```
In [84]: df.head(2)
```

```
Out[84]:
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region
0	2698	CA-2016-145317	2016-03-18	2016-03-23	Standard Class	SM-20320	Sean Miller	Home Office	United States
1	6827	CA-2018-118689	2018-10-02	2018-10-09	Standard Class	TC-20980	Tamara Chand	Corporate	United States

2 rows × 21 columns



```
In [85]: list(df.columns)
```

```
Out[85]: ['Row ID',
'Order ID',
'Order Date',
'Ship Date',
'Ship Mode',
'Customer ID',
'Customer Name',
'Segment',
'Country/Region',
'City',
'State',
'Postal Code',
'Region',
'Product ID',
'Category',
'Sub-Category',
'Product Name',
'Sales',
'Quantity',
'Discount',
'Profit']
```

```
In [86]: df1 = df[["Region", "Country/Region", "State", "City", "Sales", "Profit", "Quant
```

```
In [87]: df1.head()
```

```
Out[87]:
```

	Region	Country/Region	State	City	Sales	Profit	Quantity
0	South	United States	Florida	Jacksonville	22638.480	-1811.0784	6
1	Central	United States	Indiana	Lafayette	7499.950	8399.9760	5
2	West	United States	Washington	Seattle	3999.960	6719.9808	4
3	East	United States	New York	New York City	11199.968	3919.9888	4
4	East	United States	Delaware	Newark	499.970	5039.9856	3

```
In [91]: df1[df1["Region"] == "South"]["Sales"].sum() #Filtering south region Sales
```

```
Out[91]: 391721.905
```

```
In [92]: df1[df1["Region"] == "Central"]["Sales"].sum() #Filtering central region S
```

```
Out[92]: 491239.446800000003
```

```
In [93]: df1.groupby(["Region", "State"])["Sales"].sum() #Filtering region wise, sta
```

```
Out[93]:
```

Region	State	
Central	Illinois	80166.1010
	Indiana	43555.3600
	Iowa	4579.7600
	Kansas	2914.3100
	Michigan	76269.6140
	Minnesota	29863.1500
	Missouri	22205.1500
	Nebraska	7464.9300
	North Dakota	919.9100
	Oklahoma	19683.3900
	South Dakota	1315.5600
	Texas	170187.6018
	Wisconsin	32114.6100
East	Connecticut	13384.3570
	Delaware	17451.0690
	District of Columbia	2865.0200
	Maine	1270.5300
	Maryland	23705.5230
	Massachusetts	28634.4340
	New Hampshire	7292.5240
	New Jersey	35764.3120
	New York	310876.2710
	Ohio	78258.1360
	Pennsylvania	116511.9140
	Rhode Island	22627.9560
	Vermont	8929.3700
South	West Virginia	1209.8240
	Alabama	19510.6400
	Arkansas	11678.1300
	Florida	89473.7080
	Georgia	49095.8400
	Kentucky	36591.7500
	Louisiana	9217.0300
	Mississippi	10771.3400
	North Carolina	55603.1640
	South Carolina	8481.7100
	Tennessee	30661.8730
	Virginia	70636.7200
West	Arizona	35282.0010
	California	457687.6315
	Colorado	32108.1180
	Idaho	4382.4860
	Montana	5589.3520
	Nevada	16729.1020
	New Mexico	4783.5220
	Oregon	17431.1500
	Utah	11220.0560
	Washington	128641.2700
	Wyoming	1603.1360

Name: Sales, dtype: float64

```
In [95]: df1.groupby(["Region", "State", "City"])[["Sales"]].sum().head(10) #Filterin
```



Out[95]:

		Sales
Region	State	City
Central	Illinois	Arlington Heights
		14.112
		Aurora
		7572.968
		Bloomington
		963.940
		Bolingbrook
		218.308
		Buffalo Grove
		831.399
		Carol Stream
		1305.810
		Champaign
		151.960
		Chicago
		48539.541
		Danville
		43.372
		Decatur
		3169.473

```
In [97]: df1.groupby(["Region", "State"])[["Sales", "Profit"]].sum() #Filtering region
```

Out[97]:

		Sales	Profit
Region	State		
Central	Illinois	80166.1010	-12607.8870
	Indiana	43555.3600	18382.9363
	Iowa	4579.7600	1183.8119
	Kansas	2914.3100	836.4435
	Michigan	76269.6140	24463.1876
	Minnesota	29863.1500	10823.1874
	Missouri	22205.1500	6436.2105
	Nebraska	7464.9300	2037.0942
	North Dakota	919.9100	230.1497
	Oklahoma	19683.3900	4853.9560
	South Dakota	1315.5600	394.8283
	Texas	170187.6018	-25728.2463
	Wisconsin	32114.6100	8401.8004
	Connecticut	13384.3570	3511.4918
East	Delaware	17451.0690	9977.3748
	District of Columbia	2865.0200	1059.5893
	Maine	1270.5300	454.4862
	Maryland	23705.5230	7031.1788
	Massachusetts	28634.4340	6785.5016
	New Hampshire	7292.5240	1706.5028
	New Jersey	35764.3120	9772.9138
	New York	310876.2710	74038.5486
	Ohio	78258.1360	-16971.3766
	Pennsylvania	116511.9140	-15559.9603
	Rhode Island	22627.9560	7285.6293
	Vermont	8929.3700	2244.9783
	West Virginia	1209.8240	185.9216
South	Alabama	19510.6400	5786.8253
	Arkansas	11678.1300	4008.6871
	Florida	89473.7080	-3399.3017
	Georgia	49095.8400	16250.0433
	Kentucky	36591.7500	11199.6966
	Louisiana	9217.0300	2196.1023
	Mississippi	10771.3400	3172.9762
	North Carolina	55603.1640	-7490.9122
	South Carolina	8481.7100	1769.0566
	Tennessee	30661.8730	-5341.6936
	Virginia	70636.7200	18597.9504

		Sales	Profit
Region	State		
West	Arizona	35282.0010	-3427.9246
	California	457687.6315	76381.3871
	Colorado	32108.1180	-6527.8579
	Idaho	4382.4860	826.7231
	Montana	5589.3520	1833.3285
	Nevada	16729.1020	3316.7659
	New Mexico	4783.5220	1157.1161
	Oregon	17431.1500	-1190.4705
	Utah	11220.0560	2546.5335
	Washington	128641.2700	33402.6517
	Wyoming	1603.1360	100.1960